

# Architecture and Data Audit

## *CIND 820 Big Data Analytics Project: Milestone 2 Interim Report*

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## 2.0 From Plan to Practice

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This chapter moves from the plan set out in Milestone 1 into the architecture and data audit work itself, testing whether the design holds up against real data, and whether the methodological choices made for this project are defensible.

This chapter has four parts. First, the methodological rationale sets out why descriptive EDA is the right tool for this project, and what a simpler baseline approach would miss. Second, the data profiling and audit section presents the work completed on the PHO dataset, with the actual numbers the data produced. Third, the architecture section revisits the pipeline from Milestone 1 and shows how the pipeline logic holds together end to end. Fourth, the reproducibility section sets out the GitHub structure, the README, and how the privacy and AI commitments from Milestone 1 carry forward into this stage.

## 2.1 Methodological Rationale

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### 2.1.1 Why Descriptive EDA, and Why Now

Before any chart is produced, a harder question needs answering first: why this method, for this data, in service of these questions? It would be easy to jump straight into building dashboards. But the literature on exploratory data analysis is clear that EDA isn't a warm-up exercise before the "real" analysis. It is the real analysis, at this stage. Since Tukey's original framing of EDA in 1977, it has been treated as the standard first step in any research analysis, precisely because it surfaces distribution, outliers, and structure before any formal modelling assumptions get imposed on the data (NCBI Bookshelf, 2016). For a project whose research questions are themselves descriptive (where is the burden highest, how has it changed, where does it overlap with access gaps), EDA isn't a precursor to the method. It is the method.

There's a second reason EDA is the right starting point here, and it's specific to this dataset. The PHO Chronic Disease Incidence and Prevalence Snapshot is built on ICES-linked administrative health data, not survey data. A study comparing administrative data with self-reported survey data for chronic disease ascertainment in Ontario found that no single method has emerged as a gold standard, and that administrative data, while not collected for surveillance purposes, is a relatively inexpensive way to study large populations over time, with the caveat that some cases will always be missed (Public Health Agency of Canada study, Ontario, n.d.). That caveat is exactly why profiling the data carefully, before drawing any regional conclusions, isn't optional. What this data can and can't show needs to be established before anything is built on top of it.

One scoping note: this project is original analysis, not a replication of an existing study, so the RESER replication reporting requirements (Carver, 2010) do not apply here. Every methodological choice in this chapter is justified against this project's own dataset and research questions, not against a prior study's design.

## Gap Analysis Summary

Sections 2.1.2 and 2.1.3 work through this argument in full, but here is the shape of it: the standard baseline in this domain is a single-indicator regional ranking (section 2.1.2 identifies this and shows, using the project's own correlation results, where it breaks down). Section 2.1.3 then justifies a composite, multi-indicator burden score as the better fit for this dataset and its constraints, while being explicit, using the THIS Institute (2023) critique, about what a composite score can't do on its own. That identify-critique-justify chain is what the rest of section 2.1 builds on.

### 2.1.2 A Simple Baseline, and Why It Falls Short

The simplest possible baseline for this project would be a single-indicator ranking: take one chronic disease indicator, say, diabetes prevalence, rank Ontario's 37 public health units from highest to lowest, and call the top few “high burden” regions. It's fast, it's transparent, and anyone can reproduce it in one line of pandas.

It's also exactly the kind of approach that the literature on health system performance measurement warns against using in isolation. A single indicator tells you about one disease, in one year, with no sense of whether that region's burden is part of a broader pattern or an outlier. The EDA conducted on the PHO dataset makes this concrete: when the four chronic disease prevalence rates across PHUs in 2023 were correlated, diabetes and hypertension moved together ( $r = 0.49$ ), and asthma and COPD moved together ( $r = 0.38$ ), but diabetes and COPD were essentially uncorrelated ( $r = -0.17$ ). A region could rank near the top on one indicator and near the bottom on another. A single-indicator baseline would miss that entirely, and it would miss it in a way that directly matters for RQ1 and RQ3.

The planned method goes one step further: a composite burden score built by normalising each of the four prevalence indicators to a 0–1 scale and averaging across them, alongside the underlying indicators themselves and the PHO-provided significance flags (whether each PHU's rate is statistically higher, lower, or not different from the Ontario average). That produces a regional ranking that reflects overall burden, not just one disease's snapshot.

### 2.1.3 Being Honest About What a Composite Score Can and Can't Do

It is worth being upfront about something here, because the literature is direct about it. Composite indicators have a well-documented set of problems: a lack of transparency in how they're built, weaknesses in standardisation and weighting, and a tendency to obscure the statistical uncertainty behind a single number (THIS Institute, 2023). The same review argues that the fix isn't to abandon composite measures. It's to be transparent about how they're constructed and to present them alongside the indicators that feed them, not as a replacement for those indicators.

That is exactly how the composite burden score is used in this project. It is an equally-weighted average of four normalised prevalence rates, asthma, COPD, diabetes, and hypertension, and equal weighting is chosen deliberately, because there is no policy mandate that would justify weighting one chronic disease above another for this analysis. The composite score is a way to get a first-pass regional ranking, not a final verdict. Every region discussed in the findings is shown with its underlying indicator values alongside the composite, so a reader can see exactly

what's driving the ranking and judge for themselves whether the composite is doing the indicator justice.

### **2.1.4 Grounding RQ3 in the Real Policy Context**

RQ3 asks whether high-burden regions also face the longest wait times, in other words, whether burden and access compound each other. This isn't a hypothetical concern. Since 2018, Ontario's Public Health Standards have required all public health units to address health equity through a dedicated Health Equity Standard, covering assessment and reporting, modification of interventions, multi-sectoral collaboration, and formal health equity analysis (PMC11559488, 2024). A qualitative study of how 27 of Ontario's 34 PHUs implemented that standard found genuine appetite for exactly this kind of regional, data-driven equity analysis, but also found that PHUs frequently lack the timely, local-level data needed to do it well. Health Quality Ontario's own reporting on Northern Ontario found that the roughly 800,000 people living there are more likely to have worse health, poorer access to care, and shorter life expectancy than people in other parts of the province (HQO, n.d.). RQ3 sits squarely inside a gap that the system itself has already identified.

### **2.1.5 What the CIHI Side of the Methodology Looks Like**

On the access side, the CIHI Wait Times for Priority Procedures dataset uses a pan-Canadian definition developed by the Comparable Indicators of Access Sub Committee in 2005, adopted by federal, provincial, and territorial ministries, and validated through CIHI's ongoing Data and Information Quality Program (CIHI, n.d.). For most procedures, CIHI calculates national estimates as a weighted average of provincial submissions, with weights based on submitted surgical volumes. For this project, the key methodological point is that this is aggregate, validated, regionally comparable data, which is exactly the kind of dataset that can be joined against the PHO burden data without needing to make heroic assumptions about comparability.

### **2.1.6 Indicator Validity, Standing on Prior Work**

This is not the first application of ICES-derived administrative data to study chronic disease patterns in Ontario, and that matters for how much confidence can be placed in the PHO dataset's indicators. Gershon et al. (2021) used the same kind of administrative data infrastructure to validate COPD surveillance in Ontario, and Rosella et al. (2016) used a population-based administrative cohort to quantify the healthcare cost impact of diabetes. Both studies treat this data source as fit for population-level surveillance and cost analysis, which is precisely the level at which this project operates. The aim is not to diagnose individuals or make clinical claims, but to work at the same population, regional level where this data has already been shown to hold up.

## 2.2 Data Profiling and Audit

### 2.2.1 PHO Dataset, Findings from the Profiling Pass

This is where the project moves from a plan to evidence. A full exploratory profiling pass was run on the PHO Chronic Disease Incidence and Prevalence Snapshot, and the dataset held up well on some fronts and revealed real structure on others.

The raw file contains 39,220 rows across 8 chronic disease indicators (incidence and prevalence of asthma, COPD, diabetes, and hypertension), 37 public health units, 10 years (2014–2023), and 14 different measure types (age-standardized rates, crude rates, age-specific rates, and standardized incidence ratios, each broken out by sex). The analysis filtered down to the age-standardized rate for both sexes, which is the measure that allows fair comparison across PHUs with very different population structures. That filter left 2,960 rows, 370 per indicator across 8 indicators, exactly as expected for 37 PHUs over 10 years.

#### Completeness and suppression

Data completeness was excellent. Of the 39,220 raw rows, only 7 were flagged for suppression under PHO's data-sharing agreement, which applies when case counts fall between 1 and 4. That's a suppression rate of 0.018 percent, and every one of those suppressed rows was in the asthma incidence indicator. These 7 rows were excluded from the analytical subset. Five other columns, Rate, the 95% confidence interval, Cases, and Population, had exactly those same 7 missing values, which confirms they're tied to the same suppressed records rather than representing a separate data quality issue.

#### Descriptive statistics by indicator

The table below shows the mean, median, standard deviation, minimum, and maximum age-standardized rate (per 100,000) for each indicator across all 37 PHUs and 10 years.

| Indicator           | N   | Mean     | Median   | Std Dev | Min / Max          |
|---------------------|-----|----------|----------|---------|--------------------|
| Asthma Incidence    | 370 | 346.1    | 344.5    | 96.2    | 105.3 / 625.8      |
| Asthma Prevalence   | 370 | 14,972.9 | 14,779.6 | 1,768.1 | 9,707.0 / 19,021.0 |
| COPD Incidence      | 370 | 565.0    | 535.8    | 201.3   | 223.6 / 1,357.4    |
| COPD Prevalence     | 370 | 8,777.1  | 8,707.0  | 2,046.6 | 4,369.2 / 12,628.3 |
| Diabetes Incidence  | 370 | 800.3    | 781.0    | 137.5   | 483.3 / 1,492.3    |
| Diabetes Prevalence | 370 | 11,437.9 | 11,430.8 | 1,185.2 | 9,113.9 / 16,148.9 |

|                         |     |          |          |         |                     |
|-------------------------|-----|----------|----------|---------|---------------------|
| Hypertension Incidence  | 370 | 1,998.4  | 1,977.6  | 329.1   | 1,213.3 / 2,978.0   |
| Hypertension Prevalence | 370 | 25,751.6 | 25,837.2 | 1,695.8 | 21,222.7 / 30,300.1 |

Two things stand out immediately. First, hypertension prevalence is the dominant burden indicator by a wide margin, a mean rate of roughly 25,752 per 100,000, meaning a substantial share of the adult population in every PHU carries a hypertension diagnosis. Second, COPD prevalence has the highest relative variability of the four prevalence indicators (standard deviation of 2,046.6 against a mean of 8,777.1), indicating that COPD is where regional inequality is most pronounced. Some PHUs are dealing with a very different COPD reality than others.

### Trends over time, 2014–2023

Looking at the Ontario-wide median rate for each prevalence indicator across the full 10-year window, all four conditions show a visible dip around 2020–2021, which lines up with the COVID-19 period and almost certainly reflects reduced healthcare contact and diagnosis during that time rather than an actual drop in underlying disease prevalence. This is an important caveat carried into the limitations discussion: 2020 and 2021 data points should be read as under-counts, not as genuine improvements.

### Regional burden, 2023 snapshot

To get a single-year regional picture, a normalised burden score was built for 2023 by scaling each prevalence indicator to a 0–1 range across PHUs and averaging the four. The five highest-burden regions were Public Health Sudbury and Districts (0.80), Chatham-Kent (0.78), Durham Region (0.74), Grand Erie (0.70), and Windsor-Essex (0.70). The five lowest were Wellington-Dufferin-Guelph (0.30), Region of Waterloo (0.29), Huron Perth (0.29), Northwestern (0.27), and Ottawa (0.19).

Durham Region deserves a specific call-out, because it shows up differently depending on which lens is applied. Looking at PHO's own significance flags, which indicate whether a PHU's rate is statistically higher, lower, or not different from the Ontario average, Durham Region had 7 of its 8 indicators flagged as significantly higher than Ontario in 2023, and none flagged as lower. That's the strongest single-region pattern in the entire dataset. Ottawa sits at the opposite extreme: 7 of 8 indicators significantly lower than Ontario, and 1 higher.

### Correlation structure

The correlation matrix across the four 2023 prevalence indicators, by PHU, showed hypertension correlated positively with all three other conditions: diabetes ( $r = 0.49$ ), COPD ( $r = 0.43$ ), and asthma ( $r = 0.45$ ), which makes hypertension look like something close to a common thread across chronic disease burden in this dataset. Diabetes, by contrast, was essentially uncorrelated with COPD ( $r = -0.17$ ) and asthma ( $r = -0.08$ ). This matters for how the composite burden score is interpreted: it is not the case that “high burden” means uniformly high across all four conditions everywhere. Some regions will be driven primarily by their hypertension and



diabetes numbers, others by respiratory conditions, and the composite score is averaging across what are, in some cases, genuinely different underlying patterns.

## 2.2.2 CIHI Dataset, Findings, and a Structural Discovery

The CIHI Wait Times for Priority Procedures 2025 data tables were downloaded directly from CIHI's public portal. The file contains 18,956 provincial, national, and regional rows across 11 provinces, 14 procedures, and four metrics (50th percentile wait, 90th percentile wait, volume, and percent meeting benchmark), with 23 percent of result values missing, mostly for procedures or metrics that specific provinces don't report (for example, Quebec and most Atlantic provinces don't report a hip fracture emergency/inpatient split).

Looking at Ontario's provincial trend for five key procedures from 2014 to 2024, the same COVID-era disruption shows up here that appeared in the PHO data, but far more dramatically. Median wait time for knee replacement jumped from 85 days in 2019 to 163 days in 2020, almost double overnight. Hip replacement went from 77 to 141 days and cataract surgery from 63 to 152 days over the same single year. By 2024, hip and knee replacement and cataract surgery had partially recovered but remained above their 2019 levels, while MRI wait times kept climbing every year after 2020, reaching 46 days in 2024, the highest point in the entire series.

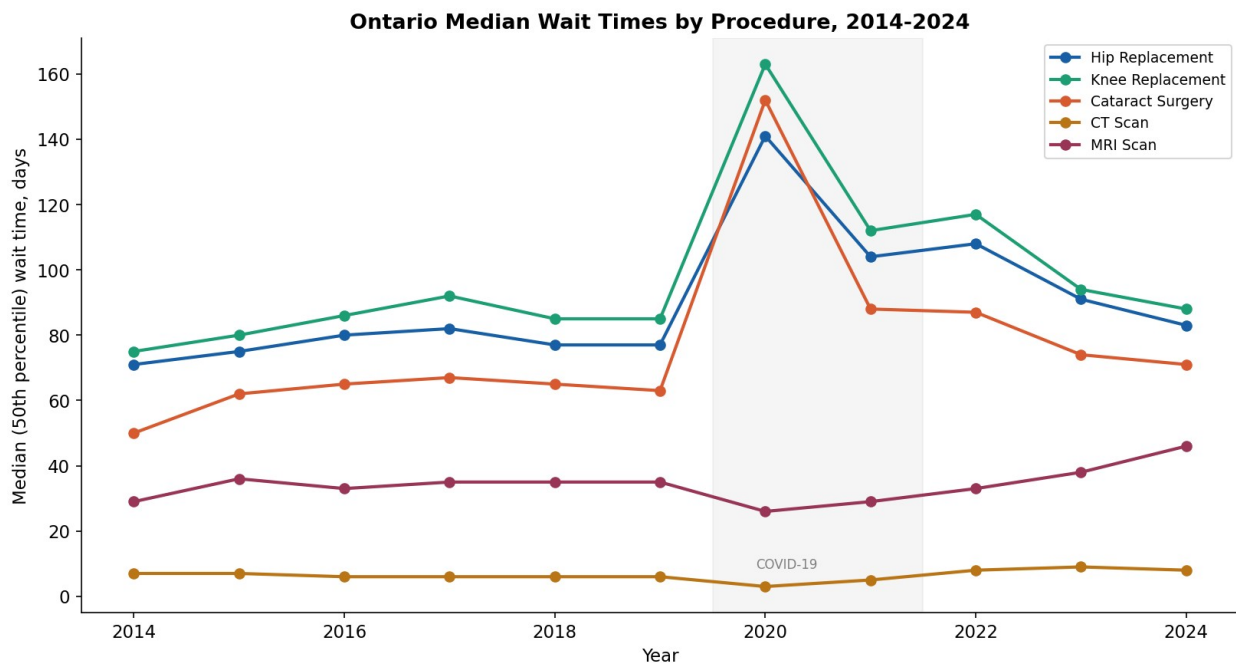


Figure 2.2, Ontario median wait times by procedure, 2014-2024

This finding changes how RQ3 needs to be approached. CIHI's 'Regional' reporting level for Ontario applies to only 2 of the 14 procedures, hip replacement and knee replacement, and uses just 6 broad health regions: Central, East, North East, North West, Toronto, and West. That is nowhere near PHO's 37 public health unit geographies. Milestone 1 had flagged that sub-provincial granularity varies by procedure and year, but the scale of that gap was not yet apparent at that stage.



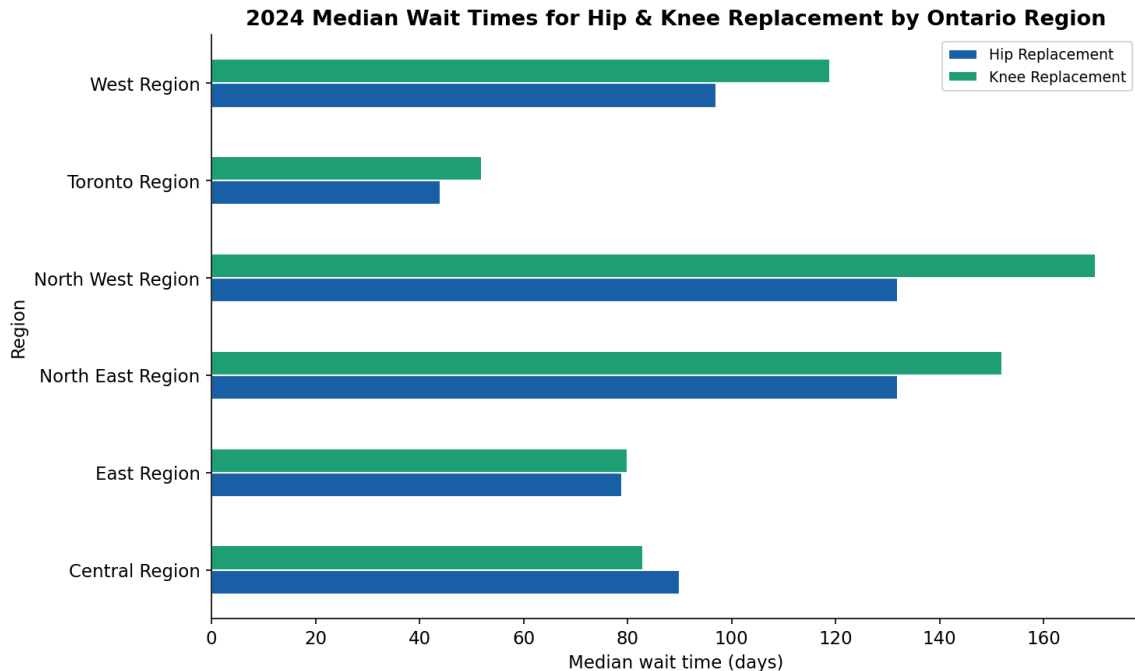


Figure 2.3, 2024 median wait times for hip and knee replacement, by Ontario region

This is where the audit produced its most useful result. Returning to the PHO file to consider how this gap could be bridged, it became clear that PHO's 'Geography' column does not only contain the 28 individual public health units that had initially been treated as "all 37 PHUs." It also contains 7 regional rollup rows: Central East, Central West, East, North East, North West, South West, and Toronto, plus an Ontario provincial total. Those 7 PHO regions map almost one-to-one onto CIHI's 6 regions. The only adjustment needed is combining PHO's Central East and Central West into CIHI's single Central Region, and treating PHO's South West as CIHI's West Region. Everything else lines up directly.

That mapping is the join key for RQ3. It is a coarser geography than anticipated in Milestone 1, 6 regions instead of 37 PHUs, but it is a real, defensible join rather than a forced one, and PHO had been providing exactly the rollup needed all along.

### 2.2.3 RQ3 Preliminary Test, Burden Meets Access

With that mapping in hand, a preliminary RQ3 test was constructed. It uses the same composite burden score methodology from section 2.2.1: normalised 2023 prevalence rates for asthma, COPD, diabetes, and hypertension are averaged. This score is computed for PHO's 7 regional rollups, then aggregated so Central East and Central West form a single Central Region score. On the access side, CIHI's 2024 median wait times for hip and knee replacement, by region, are averaged into a single access measure.

The result, across Ontario's 6 health regions:

| Region | Composite Burden Score (2023) | Avg Hip+Knee Wait, Days (2024) |
|--------|-------------------------------|--------------------------------|
|--------|-------------------------------|--------------------------------|

|                   |      |       |
|-------------------|------|-------|
| North East Region | 0.88 | 142.0 |
| Central Region    | 0.58 | 86.5  |
| West Region       | 0.56 | 108.0 |
| North West Region | 0.43 | 151.0 |
| East Region       | 0.32 | 79.5  |
| Toronto Region    | 0.31 | 48.0  |

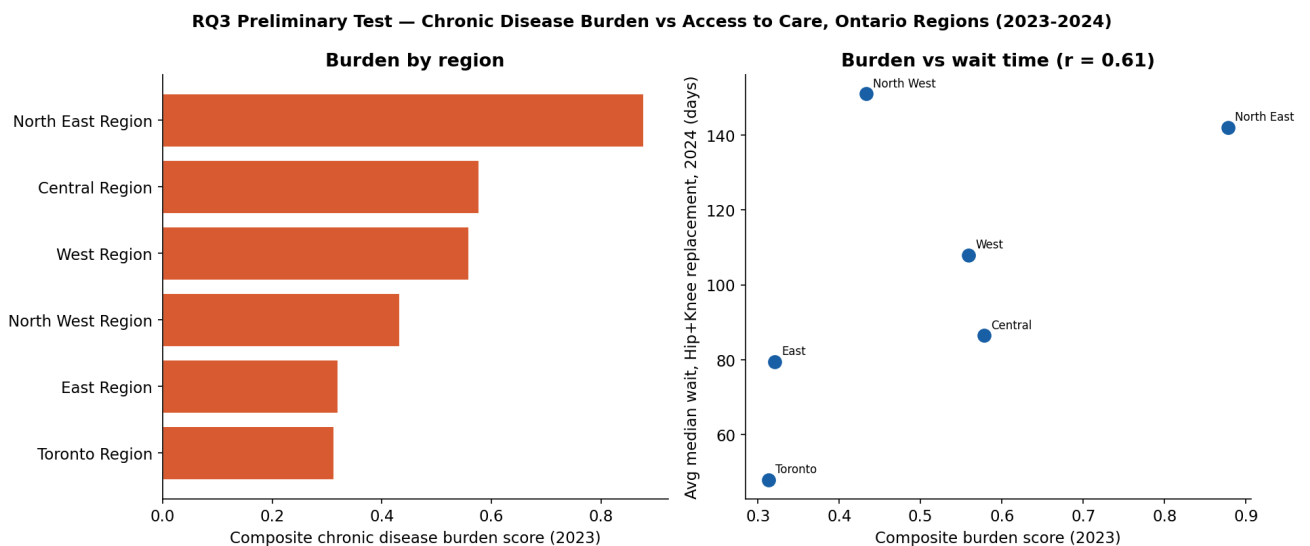


Figure 2.4, RQ3 preliminary test: chronic disease burden vs. hip/knee replacement wait times, by Ontario region

The Pearson correlation between composite burden and average wait time across these 6 regions is  $r = 0.61$ . North East Region stands out on both axes: the highest chronic disease burden in the province (0.88, noticeably above every other region) and the second-longest hip and knee replacement wait (142 days). Toronto Region sits at the opposite corner on both measures, lowest burden (0.31) and shortest wait (48 days).

It is important to be careful about what this does and doesn't show. With only 6 regions, this is nowhere near enough data points for a statistically significant correlation.  $r = 0.61$  on  $n = 6$  is a pattern worth investigating further, not a finding that should be presented to Ontario Health as evidence. North West Region is the clearest exception to the pattern: it has the second-lowest burden score (0.43) but the longest wait time in the province (151 days), which on its own tells a different story: access constraints in the north may be driven by geography and service availability rather than by disease burden. That's exactly the kind of nuance a 6-region analysis can surface but can't resolve. Going into Milestone 3, the plan is to test whether this pattern holds across multiple years rather than just the 2023/2024 snapshot, which would effectively increase the sample size and allow a more defensible statement about whether burden and access genuinely move together in Ontario.

## 2.2.4 Data Structure and Preparation

### Data Structure Classification (CIND110)

Both datasets are structured data: rectangular tables with a fixed schema, consistent column types, and one observation per row. The PHO file is a single Excel sheet in long format with 10 columns (Indicator, Measure, Year, Geography, Rate, 95% CI, Cases, Population, Significance, Suppression Flag). The CIHI file is a single Excel sheet, also long format, with 8 columns (Reporting level, Province, Region, Indicator, Metric, Data year, Unit of measurement, Indicator result). Neither dataset contains nested fields, free text, JSON, or any structure that requires parsing or flattening. There is no semi-structured or unstructured component to this project. Given that, the CIND110 work here is about reshaping (long to wide, pivot, filter) rather than parsing.

Neither dataset is large enough to require a representative sub-population. PHO has 39,220 rows and CIHI has 18,956 rows after filtering to provincial, national, and regional levels. Both are small enough to load and process in full in memory, so every figure and table in this chapter uses the complete dataset, not a sample.

### Preparation Pipeline

Both datasets arrive in long format: one row per combination of indicator, measure, year, and geography (PHO) or indicator, metric, year, province, and region (CIHI). The preparation pipeline does the following, in order:

- Filter PHO to Measure = 'Age-standardized rate (both sexes)', which is the only measure type that supports fair cross-region comparison.
- Drop the 7 suppressed PHO records, identified via the suppression flag column.
- Map the 8 raw PHO indicator names to short, consistent labels (e.g., 'Incidence of COPD in adults 20+' becomes 'COPD Incidence') for cleaner charting and tables.
- Filter CIHI to Reporting level in {Provincial, National, Regional}, excluding the Notes and footnote rows that share columns with the data rows in the raw file.
- Pivot both datasets from long to wide format for the correlation, burden-score, and join work: rows become Geography/Region, columns become each indicator, values are the relevant rate or wait-time metric for a given year.
- Normalise each PHO prevalence indicator to a 0-1 scale (min-max normalisation) before averaging into the composite burden score, so that indicators with very different rate magnitudes (asthma prevalence around 15,000 vs. COPD incidence around 565) don't distort the average.
- Apply the 7-to-6 region mapping documented in section 2.2.2 (PHO Central East + Central West -> CIHI Central Region; PHO South West -> CIHI West Region; all others map directly) to join the two datasets for RQ3.

### Integration Strategy, the Join Itself

Once both wide-format frames have a 'CIHI\_Region' column (PHO's via the mapping dictionary, CIHI's natively), the integration is a standard pandas merge: `pho_burden_df.merge(cihi_wait_df, on='CIHI_Region', how='inner')`. Central East and Central West both map to 'Central Region', so before the merge the PHO frame is grouped by CIHI\_Region and the mean of the burden score

is taken, effectively a many-to-one merge resolved by aggregation on the PHO side first. 'CIHI\_Region' is the only key; there is no multi-column key because once both frames are aggregated to the 6-region level, region name is sufficient to uniquely identify each row.

This mapping dictionary and merge logic are implemented in 03\_pho\_cihi\_join\_rq3.py in the GitHub repository, not hardcoded inline within a larger function, so they can be reviewed, corrected, or extended independently of the analysis logic.

## 2.2.5 Data Limitations and Their Impact on Claims

Milestone 1 listed the high-level limitations of both datasets. Having now profiled the PHO data directly, this section is more specific about what each limitation means for the claims that can and cannot be made.

### **Suppression (7 records, 0.018%)**

Because all 7 suppressed records are in asthma incidence and the suppression rate is negligible, this has effectively no impact on regional rankings or trend analysis. It is noted for completeness and transparency, not because it changes any conclusion.

### **COVID-era data (2020–2021)**

The visible dip in all four prevalence indicators during 2020–2021 means these years cannot be claimed to represent genuine improvements in population health. Any trend analysis that includes these years needs to either flag them explicitly as a measurement artifact or, where the research question allows it, present the trend with and without the COVID-era years to show both views.

### **PHU-level geography only**

All PHO data is aggregated to the public health unit level. No claim can be made about variation within a PHU, for example, whether burden within Durham Region is concentrated in particular municipalities. Any recommendation arising from this analysis has to be actionable at the PHU level, because that's the finest resolution the data supports.

### **Reporting lag**

PHO indicators carry a 1–2 year reporting lag. The most recent year in the dataset, 2023, reflects conditions as they were recorded through ICES linkage processes that take time to finalize. This data cannot be claimed to reflect the current (2026) state of chronic disease burden in Ontario, only the most recent state that has been validated and published.

### **Composite score as a first-pass tool, not a final ranking**

As discussed in section 2.1.3, the composite burden score is an equally-weighted average with no policy-derived weighting scheme behind it. A claim that a region with a high composite score has elevated burden across multiple chronic disease indicators is supportable. A claim that a region ranked 5th is meaningfully “worse” than a region ranked 6th is not. The composite score is precise enough to identify clusters and outliers, but not precise enough to support fine-grained ordinal claims.

**Correlation is not causation**

The correlations identified between chronic disease indicators, particularly hypertension's relationship with diabetes, COPD, and asthma, describe co-occurrence across PHUs, not a causal mechanism. These correlations can be used to identify which regions face compounding chronic disease burden, but they cannot support a claim that one condition is driving another without evidence beyond what this dataset can provide.

**CIHI regional granularity (6 regions, 2 of 14 procedures)**

CIHI's regional reporting for Ontario covers only hip and knee replacement, at the level of 6 broad health regions. This is the binding constraint on RQ3. At the 6-region level, a claim that chronic disease burden and hip/knee replacement wait times show a moderate positive relationship in 2023/2024 is supportable. No claim can be made about wait times for the other 12 procedures at a sub-provincial level, or about RQ3 at the 37-PHU resolution that the rest of the project operates at.

**RQ3 sample size (n = 6)**

The  $r = 0.61$  correlation in section 2.2.3 is based on 6 regional data points. This can be presented as a preliminary pattern that motivates further investigation, but not as a statistically significant finding. A formal hypothesis test will not be run on it until Milestone 3, when the analysis is extended across multiple years to build a larger effective sample.

## 2.3 Architecture and Logic

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The six-stage pipeline proposed in Milestone 1 is reproduced below as a diagram, with the data sources, the evaluation checkpoint, and the feedback loop made explicit. The focus here is not just what each stage does, but how the output of one stage becomes the input the next stage depends on, because that's where a pipeline either holds together or falls apart.

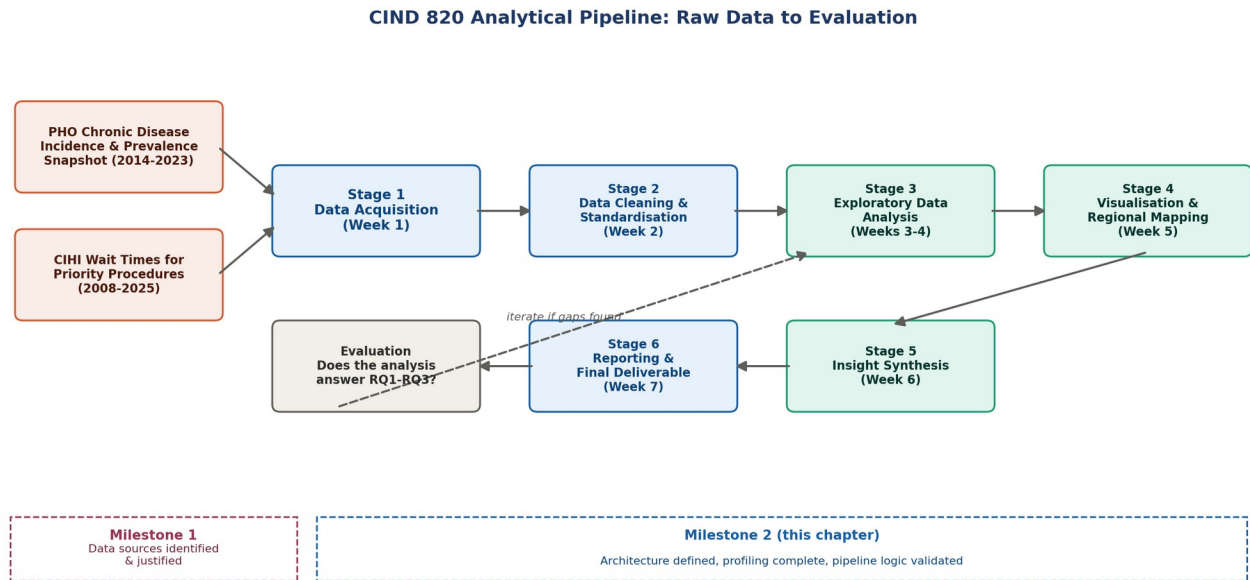


Figure 2.1, Analytical pipeline from raw data to evaluation, with feedback loop

### 2.3.1 Stage 1 to Stage 2, From Two Datasets to One Clean Frame

Stage 1 produces two raw files: the PHO Snapshot export and the CIHI XLSX data tables, each saved with a version-stamped filename and a logged source URL and access date. Stage 2 takes those two files and applies the cleaning steps in section 2.2.3, filtering, suppression handling, indicator renaming, and geography standardisation. The output of Stage 2 is not one merged file yet; it's two cleaned, analysis-ready frames that share a common geography key. That shared key is the load-bearing element of the whole pipeline. Every downstream stage depends on PHO and CIHI geographies meaning the same thing.

### 2.3.2 Stage 2 to Stage 3, From Clean Data to Evidence

Stage 3 is where this chapter's findings came from. The PHO frame feeds directly into the descriptive statistics, trend analysis, regional burden scoring, and correlation analysis in section 2.2.1. The CIHI frame feeds into the provincial trend analysis and regional wait-time breakdown in section 2.2.2. And the join logic in `03_pho_cihi_join_rq3.py`, pairing each region's burden score with its corresponding wait-time data using the 7-to-6 region mapping, is what produced the RQ3 preliminary result in section 2.2.3.

### 2.3.3 Stage 3 to Stage 4, From Numbers to Pictures

Every output from Stage 3 has a corresponding visual in Stage 4. The descriptive statistics become the summary table in section 2.2.1. The regional burden scores become the heatmap and the top/bottom region charts. The correlation matrix becomes a correlation heatmap. None

of the Stage 4 visuals are decorative. Each one exists because it's the clearest way to communicate a specific Stage 3 finding to the audience defined in Milestone 1: Ontario Health's regional planning and system performance teams.

### 2.3.4 Stage 4 to Stage 5 to Stage 6, From Pictures to Recommendations

Stage 5 is where individual charts are stepped back from to ask what they mean together. The Durham Region finding, 7 of 8 indicators significantly higher than Ontario, is a Stage 3 output, but identifying Durham as a priority region, and explaining why that matters for RQ1 and (once CIHI is joined) RQ3, is Stage 5 work. Stage 6 then takes those synthesized insights and structures them into the final report format: executive summary, narrative, visuals, limitations, and recommendations.

### 2.3.5 The Evaluation Checkpoint and Feedback Loop

The diagram includes an evaluation checkpoint after Stage 6, with a dashed feedback arrow back to Stage 3. This isn't decorative either. If, once CIHI is joined, the geography mapping doesn't hold for enough PHUs to answer RQ3 meaningfully, the appropriate response is to go back to Stage 3, not to force a weak join through to Stage 6 and write around the problem. Building that loop into the architecture now, before that problem arises, means the response is already decided in advance.

### 2.3.6 Leakage Prevention and Validation Strategy

This milestone's pipeline is descriptive, not predictive. There is no model being trained, so the classic train/test leakage problem (where information from the test set influences training) doesn't apply in its usual form. But there is an analogous risk in trend and regional analysis: temporal leakage, where a claim about 'current' conditions is quietly informed by data from periods that shouldn't count as 'current.' This is addressed directly in two places. First, the composite burden score and the RQ3 join both use 2023 PHO data and 2024 CIHI data specifically, the most recent complete year available in each dataset, rather than blending multiple years into an average that could obscure which year is actually driving a result. Second, the 2014–2023 trend lines discussed in sections 2.2.1 and 2.2.2 explicitly flag 2020–2021 as a COVID-affected anomaly rather than letting those points silently pull trend lines in a direction that doesn't reflect underlying disease burden.

Looking ahead to Milestone 3: if the project moves toward any predictive or forecasting component, for example, projecting 2024–2025 burden from the 2014–2023 series, the validation strategy will be a time-based split, not a random split. Training on earlier years and validating on the most recent 1–2 years is the only split that doesn't leak future information into the model, given that chronic disease rates are autocorrelated year over year. A random train/test split across years would let the model see 2023 during training and be tested on 2021, which would overstate performance. This is noted now, before any modelling code is written, so that the validation strategy is a design decision rather than an afterthought.

Concretely, this means scikit-learn's `TimeSeriesSplit` rather than `StratifiedKFold` or a plain random split. `TimeSeriesSplit` enforces that test indices always come after train indices and that successive training sets only grow forward in time, which is precisely the constraint documented



in scikit-learn's own cross-validation guide as the reason standard k-fold is inappropriate for time-ordered data (scikit-learn, n.d.). For a 10-year annual series like the PHO prevalence data, that would mean folds such as train on 2014–2020 / test on 2021, then train on 2014–2021 / test on 2022, and so on, an expanding-window approach that mirrors how the model would actually be used: making a forecast with only the data that existed at that point in time.

### 2.3.7 Where This Chapter Sits in the Pipeline

The bands at the bottom of Figure 2.1 mark what each milestone covers. Milestone 1 established that the data sources exist, are public, and are appropriate for the research questions. This chapter, Milestone 2, covers the architecture definition, the completed PHO profiling, and validation that the pipeline logic holds together from Stage 1 through to the evaluation checkpoint. The CIHI profiling and the RQ3 join are the work that carries this pipeline from Stage 2 into the back half of Stage 3, and they are the explicit bridge into Milestone 3.

### 2.3.8 The Modeling and Evaluation Architecture for Milestone 3

Sections 2.3.1 through 2.3.7 describe the architecture that has been built and validated for this milestone's descriptive work: two cleaned frames sharing a common geography key, a single-year composite burden score, and a preliminary correlation test for RQ3. The leakage and validation strategy set out in section 2.3.6 was deliberately written ahead of any modelling work, so that the design decision would not become an afterthought once Milestone 3 begins. Figure 2.5 sets out the six-stage architecture that strategy points toward: the pipeline planned for Milestone 3, where the project moves from a single-year descriptive snapshot to a multi-year forecasting comparison that addresses RQ2 and RQ3 directly.

#### Preprocessing for the Expanded Panel

The PHO and CIHI extraction and cleaning steps documented in sections 2.2.4, 2.3.1, and 2.3.2 already produce two analysis-ready frames for the single 2023/2024 snapshot used in this milestone. Milestone 3 extends both frames to their full longitudinal range, the 2014-2023 PHO series and the 2008-2024 CIHI series, and runs an automated profiling pass over the expanded frames before reshaping them with the same melt-to-long approach used in section 2.2.4. Expanding from one year to ten increases the chance that some PHU-indicator-year cells fall into PHO's suppression range. The single 2023 snapshot used in this milestone had only 7 such cells (0.018 percent, section 2.2.1), all excluded outright, but across a ten-year, 37-geography panel that count could be high enough that exclusion would create gaps in specific regional time series. For the expanded panel, Milestone 3 plans a spatial K-nearest-neighbours imputation ( $k=3$ ), estimating a suppressed cell from the trajectories of its geographically nearest PHUs rather than dropping the row or substituting a province-wide mean, which would flatten exactly the regional variation this project is trying to measure.

#### From Composite Burden Score to a System Pressure Score

Section 2.1.3 introduced the composite burden score as an equally-weighted average of four normalised prevalence indicators. That score speaks to RQ1: which regions carry the highest chronic disease burden. Milestone 3 extends the same normalisation logic to the access side of RQ3. Each indicator, prevalence rates and CIHI wait-time percentiles alike, is converted to a Z-score,  $Z$  equals  $(X \text{ minus the mean})$  divided by the standard deviation, which puts

measurements in percent of population and days of waiting onto the same standardised scale. Combining the two standardised components into a single System Pressure Score, SPS equals  $\alpha$  times Z-prevalence plus  $\beta$  times Z-wait-time, with  $\alpha$  and  $\beta$  both set to 0.50 as a starting, equally-weighted split, gives planners one ranked number per region per year that moves the project from 'these two things are correlated' (the RQ3 finding in section 2.2.3) toward 'this is how pressured this region is right now, combining both dimensions.' As with the composite burden score in section 2.1.3, the equal weighting is a starting point to be presented alongside its component parts, not a substitute for them.

### **Chronological Split, Scaling, and Leakage Prevention**

Section 2.3.6 established that any predictive work on this data must use a time-based split, training on earlier years and validating on the most recent ones, because chronic disease rates are autocorrelated year over year and a random split would let the model see the future during training. Milestone 3 implements that principle as a single chronological split across the ten-year PHO series: the 2014-2021 period forms the training partition, and 2022-2023 is held out as validation, an 8-year training window against a 2-year validation window. Z-score scaling parameters, the mean and standard deviation used in the SPS calculation above, are fit exclusively on the 2014-2021 training partition and then applied unchanged to the 2022-2023 validation partition, exactly the TimeSeriesSplit logic described in section 2.3.6. One additional safeguard addresses circular reasoning rather than temporal leakage: raw patient volume totals are excluded from the feature set entirely, because a count of patients is close enough to the utilization rate the model is trying to forecast that including it would let the model effectively restate the outcome rather than predict it.

### **Algorithmic Comparison: OLS Baseline Against a VAR Model**

The RQ3 preliminary test in section 2.2.3 established a single correlation coefficient ( $r = 0.61$ ,  $n = 6$ ) between burden and access at one point in time. Milestone 3 turns that static relationship into a forecasting comparison. An ordinary least squares (OLS) regression serves as the baseline, treating each region's future System Pressure Score as a linear function of its own past values. Against that baseline, a multivariate vector autoregression (VAR) model treats chronic disease indicators and wait-time percentiles as co-dependent series, allowing each to inform forecasts of the other, which is closer to the compounding burden-and-access relationship RQ3 is actually asking about. Both models are evaluated on the 2022-2023 hold-out using root mean squared error (RMSE) and mean absolute error (MAE), the standard pair of scale-dependent accuracy metrics for continuous forecasts. If the VAR model outperforms the OLS baseline on both metrics, that is evidence the burden-access relationship has the kind of dynamic, feedback structure RQ3 hypothesises. If it does not, the simpler OLS baseline is the more defensible model to report.

**Planned Modeling and Evaluation Architecture (Milestone 3)**

*Figure 2.5, Planned modeling and evaluation architecture for Milestone 3*

Every design choice in Figure 2.5 is a direct extension of a decision already justified earlier in this chapter: the composite score methodology from section 2.1.3, the region mapping discovered in section 2.2.2, and the time-based validation strategy from section 2.3.6. Milestone 2's contribution is establishing that these foundations hold against real data. Milestone 3's task is to build the forecasting layer on top of them.

## 2.4 Reproducibility

Everything in sections 2.2 and 2.3 needs to be reproducible by anyone with access to the same two public datasets. That is the standard this project is held to, and it is also the standard committed to in the Milestone 1 GenAI declaration: every result traceable to specific rows or aggregations in the source data, and every piece of analysis documented in a reproducible notebook.

### 2.4.1 Repository Structure

The GitHub repository for this project is [github.com/mamkon/Architecture\\_and\\_Data\\_Audit](https://github.com/mamkon/Architecture_and_Data_Audit) and is organised as follows:

```
Architecture_and_Data_Audit/
|-- README.md
|-- LICENSE
|-- data/
|   |-- raw/
|   |   |-- PHO_Chronic_Disease_Inc_Prev_Snapshot_2014_2023.xlsx
|   |   |-- CIHI_Wait_Times_Priority_Procedures_2025.xlsx
|   |-- processed/
|-- notebooks/
|   |-- 01_pho_eda.ipynb
|   |-- 02_cihi_eda.ipynb
|   |-- 03_pho_cihi_join_rq3.ipynb
|-- outputs/
|   |-- figures/
|   |   |-- fig1_dataset_overview.png
|   |   |-- fig2_suppression_analysis.png
|   |   |-- fig3_prevalence_trends_ontario.png
|   |   |-- fig4_regional_burden_heatmap.png
|   |   |-- fig5_top_bottom_regions.png
|   |   |-- fig6_disease_correlation.png
|   |   |-- fig7_significance_distribution.png
|   |   |-- fig8_cihi_ontario_trends.png
|   |   |-- fig9_cihi_ontario_regions_2024.png
|   |   |-- fig10_cihi_benchmark_by_province.png
|   |   |-- fig11_rq3_burden_vs_access.png
|   |   |-- fig12_modeling_pipeline.png
|   |-- tables/
|   |   |-- eda_summary_stats.csv
|   |   |-- cihi_ontario_2024_summary.csv
|   |   |-- rq3_burden_vs_access.csv
|-- reports/
|   |-- CIND820_Milestone2_Interim_Report.pdf
|-- docs/
|   |-- privacy_and_ethics_check.md
|   |-- ai_use_declaration.md
|   |-- pipeline_diagram.png
-- (end)
```

## 2.4.2 README, Core Content

The README opens with the project summary from section 1.2 of Milestone 1, condensed, followed by the dataset table with source URLs, licences, and access dates, the repository structure shown above, Open in Colab links for each of the three notebooks, and a status section. Each notebook's first code cell clones the repository when running in Colab, so the relative data and output paths used throughout resolve correctly without manual setup. The current status section reads: PHO dataset fully profiled (39,220 rows, 8 indicators, 37 geographies); CIHI dataset fully profiled (18,956 rows, 14 indicators, 11 provinces); the 7-to-6 regional mapping identified and documented; RQ3 preliminary test completed ( $r = 0.61$ ,  $n = 6$ ); next step for Milestone 3 is extending the regional join across multiple years.

## 2.4.3 Privacy Check, Carried Forward

The privacy and ethics audit from Milestone 1 section 3.4 carries forward unchanged: both datasets are fully aggregated, contain no personally identifiable information or personal health information as defined under PHIPA, and are published under open government licences that explicitly permit academic use. Having now worked directly with the PHO data at the row level, this assessment is confirmed: every row examined is an aggregate count or rate at the PHU-year-indicator level, with no individual-level fields anywhere in the file.

Both datasets originate from individual-level health records before aggregation: the PHO snapshot from ICES-linked OHIP physician claims and hospital discharge records, and the CIHI wait time tables from provincial wait-list submissions for individual procedures. Because the analysis ultimately traces back to records about identifiable people, the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2) is the relevant framework for assessing whether this project requires Research Ethics Board review (Government of Canada, 2022). TCPS 2 Article 2.4 exempts secondary use of information that is already publicly available through established channels, or that has been fully aggregated such that no individual can be identified, directly or indirectly, from REB review. Both the PHO and CIHI datasets meet that description at the PHU/region-year-indicator level used throughout this project, which is why no institutional REB approval is triggered by this analysis.

## 2.4.4 AI Use, Carried Forward, With One Addition

The GenAI declaration from Milestone 1 section 5 continues to govern this chapter: AI tools were used for drafting, structuring, and clarity edits, with all analysis, code, and interpretation produced and verified by the student. One addition for this milestone: every literature source cited in section 2.1 was independently retrieved and checked against its original publication before being cited here, consistent with the hallucination-risk mitigation committed to in Milestone 1 section 5.3.

## 2.5 The Interim Report

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This chapter functions as the interim report for Milestone 2. This closing section draws together three threads that run through it: how the plan set out in Milestone 1 has been updated and validated, where the architectural decisions made in this chapter are justified, and how AI tools were used in producing it.

### 2.5.1 Milestone 1 Updates Incorporated

Milestone 1 established the three research questions, identified the PHO and CIHI datasets as the data sources, and proposed the six-stage pipeline later reproduced as Figure 2.1. Section 2.0 set out this chapter as the direct continuation of that plan, and the rest of the chapter updates it with results from real data. The PHO profiling pass (section 2.2.1) replaces the plan's assumptions with actual numbers: 37 geographies, negligible suppression, a COVID-era dip across all four conditions, hypertension as the dominant indicator, and a correlation structure that shapes how the composite burden score is interpreted. The CIHI profiling pass (section 2.2.2) carries out the comparison Milestone 1 anticipated and surfaces a structural discovery, PHO's own 7-region rollups, that resolves the geography mismatch Milestone 1 had flagged as a risk. Section 2.2.3 turns that discovery into a preliminary RQ3 result ( $r = 0.61$ ,  $n = 6$ ). Sections 2.3.1 through 2.3.7 validate that the pipeline logic from Milestone 1 holds together end to end against this real data, and section 2.3.8 extends that validated architecture into the modeling and evaluation plan for Milestone 3.

### 2.5.2 Architectural Justifications, Summarized

Every architectural decision in this chapter is justified at the point where it is made, and this section is an index to those justifications rather than a restatement of them. Descriptive EDA is justified as the right method for this project's research questions in section 2.1.1. The choice of a composite burden score over a single-indicator baseline is justified by the project's own correlation results in section 2.1.2, with the limitations of composite indicators acknowledged directly in section 2.1.3. RQ3 is grounded in Ontario Health's own published policy gaps in section 2.1.4, and the validity of both datasets is grounded in prior literature in sections 2.1.5 and 2.1.6. The region mapping that makes RQ3 possible is not assumed in advance, it is discovered during profiling and documented as such in section 2.2.2. The leakage and validation strategy in section 2.3.6 is written before any modelling work begins, specifically so that it is a design decision rather than an afterthought, and section 2.3.8 shows that strategy carried through into a concrete modeling architecture for Milestone 3.

### 2.5.3 AI Use Declaration

Claude (Anthropic) was used in producing this chapter for drafting, structuring, and clarity edits: organising section content under the required headings, rewording sentences for an impersonal academic register, formatting tables, and matching the document to the TMU report template. Claude was not used to generate analysis results, statistical findings, or interpretive claims. Every number reported in this chapter, including the descriptive statistics in section 2.2.1, the correlation coefficients ( $r = 0.49$ ,  $r = 0.38$ ,  $r = -0.17$ , and the RQ3 result  $r = 0.61$ ), the 37-geography count, and the 7-record suppression total, comes from the student's own pandas-

based exploratory analysis, run independently in the notebooks listed in section 2.4.1 and verified by the student before being written up here.

Every literature source cited in section 2.1 was independently retrieved by the student and checked against its original publication before being cited, consistent with the hallucination-risk mitigation committed to in Milestone 1 section 5.3. This declaration carries forward and remains consistent with the GenAI use policy acknowledged in Milestone 1 section 5. The student can explain and justify every analytical decision, citation, and claim made in this chapter on request.

## References

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- Canadian Institute for Health Information. (n.d.). Wait time metadata.  
<https://www.cihi.ca/en/wait-time-metadata>
- Gershon, A. S., Thiruchelvam, D., Stanbrook, M. B., Hubac, J. M., Beaver, S., Loughheed, D., & To, T. (2021). Using administrative data for surveillance of chronic obstructive pulmonary disease in Ontario, Canada. *Canadian Journal of Public Health*, 112(1), 1–10.  
<https://doi.org/10.17269/s41997-020-00426-y>
- Government of Canada, Panel on Research Ethics. (2022). Tri-Council Policy Statement: Ethical conduct for research involving humans, TCPS 2 (2022).  
[https://ethics.gc.ca/eng/policy-politique\\_tcps2-eptc2\\_2022.html](https://ethics.gc.ca/eng/policy-politique_tcps2-eptc2_2022.html)
- Health Quality Ontario. (n.d.). Health equity in Northern Ontario.  
<https://www.hqontario.ca/System-Performance/Specialized-Reports/Health-Equity-in-Northern-Ontario>
- National Center for Biotechnology Information. (2016). Exploratory data analysis. In *Secondary analysis of electronic health records*. <https://www.ncbi.nlm.nih.gov/books/NBK543641/>
- Public Health Agency of Canada. (n.d.). Ascertainment of chronic diseases using population health data: a comparison of health administrative data and patient self-report.  
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3557162/>
- Rosella, L. C., Lebenbaum, M., Fitzpatrick, T., O'Reilly, D., Wang, J., Booth, G. L., & Wodchis, W. P. (2016). Impact of diabetes on healthcare costs in a population-based cohort: a cost analysis. *Diabetic Medicine*, 33(3), 395–403. <https://doi.org/10.1111/dme.12922>
- scikit-learn developers. (n.d.). 3.1. Cross-validation: evaluating estimator performance, TimeSeriesSplit. scikit-learn 1.9.0 documentation.  
[https://scikit-learn.org/stable/modules/cross\\_validation.html](https://scikit-learn.org/stable/modules/cross_validation.html)
- THIS Institute. (2023). The problem with composite indicators. The Healthcare Improvement Studies Institute. <https://www.thisinstitute.cam.ac.uk/research/outputs/the-problem-with-composite-indicators/>
- Variability in public health programming and priorities to address health inequities across public health units in Ontario, Canada. (2024). PMC.  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11559488/>